

Selected Formalisms from the Event Density Framework

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Abstract

Event Density (ED) posits a discrete substrate of *events* fixed by thirteen postulated primitives, with the irreversible *commitment* of events, the arrow of time made physical, as its keystone. Parts I and II worked the sectors where ED reproduces a law's *form* and inherits its *magnitude*. This paper works the sector that is different in kind: the results the substrate **forces**, as theorems and machine-precision structural invariants, rather than inheriting a number for. Each follows from one or more of three architectural facts about the substrate: its finiteness, its cleanness (it carries no reflection primitive, and its selection is blind to internal phase), and the irreversibility of its commitments, the arrow. **(1) Locality is exact.** The controlled information capacity between distinct substrate regions is measured to be exactly zero ($L2 = 0.000000$, bit-for-bit, with a passing local control), so the substrate's correlations are common-cause, never signals on the tested partitions; combined with the entanglement sector's reproduction of the Bell correlations (Part II), this delivers Bell locality without superdeterminism. **(2) Charge is topological.** Committed $U(1)$ polarity carries a quantized integer winding $w \in \mathbb{Z}$ ($\pi_1(U(1)) = \mathbb{Z}$, to machine precision), conserved and irreversibility-protected, obeying an integral Gauss law (circulation $= 2\pi w$, loop-independent). No charge magnitudes are produced, and the determined Coulomb field lives one coarse-graining layer up (an open edge). **(3) Parity violation is necessarily spontaneous.** The chirality operator reads in ED as $\gamma^5 = (\text{arrow}) \times (\text{spatial orientation})$; the parity-clean commitment map has $\det H(k)$ even in k , so its chirality winding is identically zero for every channel-count, the clean substrate is vector, a theorem, and any parity violation must break a symmetry rather than obey a law. Which force is chiral is inherited (the pseudoreality of $SU(2)$), not derived. **(4) Classical objectivity is unbounded**, because a committed record costs no dynamical budget: the classical *is* the committed. We are candid about the price of this purity: the sector carries no confirmed prediction (its distinctive one, parity handedness correlated with the matter/antimatter sign, is unconfirmed). And we close on a fifth theorem that cuts the other way. The same finiteness that makes locality exact **bounds** the substrate: using the integers as an external ruler, ED's finite-memory class reproduces the prime *template* exactly and provably cannot reproduce the *escape* (primality itself), a limit anchored not to ED's own grading but to a theorem outside it, and hence the program's one proven negative. Locality measured inside, and the same ceiling certified outside, are one fact seen twice.

1. The substrate, its architectural facts, and what “theorem” means here

Event Density (ED) is a substrate ontology. Beneath the continuous fields and smooth spacetime of established physics it posits a discrete, graph-like layer of *events*, and treats the familiar continuum laws as coarse-grained readings of that layer. As in Parts I and II, this paper does not argue for the substrate from first principles. It takes the substrate as given and works one sector through, stating at each step exactly what is derived and what is inherited. What makes this sector distinct is stated at the outset, because it governs everything below and it is also the sector’s honest price.

Postulated, not derived. The substrate is specified by thirteen primitives, P01–P13 (Paper_087). They are postulated, not derived from a deeper layer, and ED makes no claim that they are minimal or unique. Its epistemic standing is that of the substrate-ontology programs it sits beside, the causal-set program and ’t Hooft’s cellular-automaton interpretation, which posit a discrete substrate and argue from its downstream reach, rather than the operational-reconstruction tradition (Hardy; Chiribella–D’Ariano–Perinotti; Masanes–Müller), which derives quantum theory from operational axioms by closed proof. Every result below is a *conditional structural derivation*: given the thirteen primitives, the stated consequence follows; removing a load-bearing primitive breaks it.

The arrow. One primitive is the keystone of this paper. P11 (commitment irreversibility) is the substrate event at which a chain’s multi-channel participation collapses to a single channel, the unselected phases are randomized, and no substrate operation returns the post-commitment state to the pre-commitment one. The one-way character of the substrate, the arrow, is supplied jointly by P11, time homogeneity (P13), and the strictly retarded support of the participation kernel V1 (Theorem T18, Paper_093); ED does not claim the arrow reduces to any one of these alone. Where a popular account of ED calls commitment “the tick of time,” this is its precise content: an irreversible substrate event, forward-supported at the kernel level.

The architectural facts this sector turns on. Parts I and II leaned on the participation and bandwidth primitives. This sector rests on three architectural facts about the substrate, and every result below is a consequence of one or more of them. The arrow is the keystone that recurs; the other two are the substrate’s shape.

- **Finiteness.** The substrate is finite-reach and finite-memory: every determination is a function of a bounded local state (Paper_087). *Finite reach* is why a committed fact is strictly local, a node depends only on its neighborhood (§2); *finite memory* is why there is a structure the substrate provably cannot compute (§6). These are the paper’s two ends, and they are one fact of finiteness read at its two limits.
- **Cleanness.** The substrate carries no reflection primitive, and its selection rule (Σ) reads the event field ρ and the graph structure *only*, never the internal U(1) phase supplied by P09. The phase-blindness makes the topological charge quantized but inert (§3); the absence of any reflection makes the emergent rules parity-symmetric, which is what forces the vector/parity theorem (§4). Two faces of the substrate carrying no structure it was not given.
- **Irreversible commitment, the arrow.** P11 again, now in its structural role: a committed fact cannot be un-made. This is the keystone, and it threads the middle of the paper, it protects the charge winding from slipping (§3), it supplies the arrow factor of the chirality operator (§4), and it is why a classical record costs nothing to keep (§5).

All three are themselves postulated, part of the thirteen, not derived. The claim of this paper is not that they are inevitable but that, granted them, a specific family of results follows *as theorems*, not as forms with an inherited magnitude.

Form-derived, value-inherited, and the price of this sector. ED separates the *form* of a law (its functional shape and structural constants) from its *magnitude* (frequently inherited from measurement); the separation governed every claim in Parts I and II (Paper_095). It applies here too, but the balance shifts, and the shift is the point. In the gravity sector the forms were derived and the magnitudes of G and a_0 inherited, and the sector earned its keep with a falsifiable, confirmed prediction (the acceleration scale tracks the cosmic expansion rate). In this sector the results are firmer in *kind*, genuine theorems and machine-precision invariants rather than magnitude-inheriting forms, but the sector carries **no confirmed prediction**. Its one distinctive forward bet, a spontaneous parity handedness correlated with the matter/antimatter sign (§4), is unconfirmed and not near-term testable. We say this now, and again in §7, so that the density of theorems below is not misread: Part III trades an empirical scalp for derivation purity. It is a real trade, in both directions.

A calibration on the word “theorem.” Because this sector’s claim is precisely that its results are firmer in kind, we fix the vocabulary up front, and we hold to it. Three strengths appear below, and they are not the same:

1. an **analytic theorem**, a short mathematical proof from stated inputs, of which the paper has one: the even-determinant result that the clean substrate is vector (§4). Its close cousin is a **structural derivation**, a forced consequence of named prior results rather than a fresh proof (the accounting theorem that committed records cost no budget, §5, tiered Derived-structural, not an analytic proof);
2. a **machine-precision measured invariant carrying a derived protection**, a result read off the certified substrate to exact numerical precision, with a primitive guaranteeing it cannot be otherwise (the exactly-zero channel capacity, §2; the integer winding, conserved by P11, §3);
3. a **theorem-anchored proven limit**, a negative result whose “cannot” is guaranteed by mathematics external to ED (the finite-memory ceiling, §6).

All three are firmer than the Part I/II baseline, “form-forced on postulates, magnitude inherited.” That is the sector’s claim, and its ceiling. We never blur a measured exact-zero into “we proved,” and we mark which of the three each result is.

Coarse-graining. Finally, as before, the continuum objects that appear here (a Coulomb field, a Dirac sector, a classical record) are not substrate primitives but coarse-grained readings of substrate participation, obtained when the scale separation holds (the direct coarse-graining transport, DCGT). Two of this paper’s honest open edges, the Coulomb field of §3 and the relativistic Dirac descent of §4, live at exactly that coarse-grained layer, and we mark them where they bite. With that established, the sector proceeds: exact locality (§2), topological charge (§3), the vector/parity theorem (§4), free objectivity (§5), and the one proven limit (§6).

2. Locality is exact: common cause, not channel

Begin with the property the whole sector rests on, and the one ED can show most directly: that a committed fact is strictly local. The test is a coding experiment on the certified substrate, and its result is not a bound or an estimate but an exact zero.

The setup. Partition the certified substrate into regions: the two halves of one stratum (call them A-left and A-right), and a region B across the substrate’s decoupling boundary. Encode a K -ary message by setting A-left’s initial condition to one of K distinct values, evolve the substrate to completion, and try to recover the message from a distant region, A-right within the stratum, and B across the boundary. Channel *capacity*, the supremum of recoverable information over all encodings and readouts, is the observable-independent measure of whether any signal passes.

The result.

Changing the encoded message ($0.0 \rightarrow 0.45$) leaves the distant region’s final state **byte-identical**, $L2 = 0.000000$, across every baseline tested (5/5), while the local-response control passes: a region *does* respond to its own initial condition ($L2 = 1.8$). The message does not propagate to the distant region at all. Not weakly. Exactly zero (Paper A1).

Tier: the exact zero is Measured (a direct byte-level diagnostic, not an estimate); the controlled channel capacity between distinct regions is Derived from it. The logic is immediate: if no perturbation to the source region produces any change in the distant region, the mutual information between input and output is identically zero, so its supremum over all encodings, the capacity, is zero. A committed node depends only on its local neighborhood (the adjacency and transport primitives P02–P05); there is no controllable channel to have a capacity. The control is what certifies the substrate is working, a region responds to local input and is exactly insensitive to distant input, which is locality stated as a measurement.

What the zero means: common cause, not channel. A companion measurement had found a nonzero within-stratum correlation, $\Delta > 0$, and one might read it as a faint signal. The exact-zero capacity rules that out and reframes it. The within-stratum correlation is **shared-origin co-variation**: within one run, both halves of a stratum share the run’s seed and the front’s passage, so across an ensemble they co-vary, with no information flowing between them. Across the boundary, independent seeds give independent origins and no correlation. So the boundary is an **observational (common-cause) structure, not a transmission channel**, and its “severance” is severance of shared origin, not of a signal path, because there is no signal path anywhere. This is the textbook hallmark of common cause, nonzero observational mutual information alongside exactly zero interventional mutual information, the two rungs of Pearl’s causal hierarchy coming apart as far as they can.

The foundational payoff: locality without superdeterminism. Put the three facts together, local commitment, exact capacity-zero, and outcomes that are stochastic at the moment of commitment, and ED accounts for the Bell/EPR correlations with no signal channel anywhere *and* no hidden-variable conspiracy. One attribution matters here, and it is a division of labor: the Bell–EPR *correlations themselves* are reproduced in the entanglement sector, which recovers the Bell–Tsirelson structure exactly while inheriting the Tsirelson bound and adding no new bound (Part II); what A1 supplies is the exact no-signaling that makes that reproduc-

tion *local*. With both in hand, ED holds a genuine advantage over two standard alternatives. A superdeterministic account buys locality by correlating the measurement settings with the hidden state, a conspiracy at preparation; a signaling-collapse account buys the correlations by letting one wing influence the other. ED needs neither: the correlations are common-cause (set at a shared origin), the no-signaling is exact (capacity zero), and the arrow is forward. The nonlocal-looking correlation and the strict locality of every committed fact are not in tension, they are the observational and interventional faces of one structure.

What this is, and is not. The result is an exact zero on the cases tested, one certified substrate and five baselines with a passing control, not a proof for every possible region geometry, though ED’s locality makes a nonzero result on any *local* partition structurally surprising. And it is an *empirical* confirmation of ED’s locality on the substrate, not a derivation of locality from the contrast-first ontology; ED’s “no unmediated nonlocal influence” is shown here as exact channel-zero, not deduced from a deeper principle. One honesty note travels with it, because it is instructive: the first pass used a k-nearest-neighbor decoder whose mutual-information estimate is positively biased and read a spurious ~ 2.8 bits even across the decoupled boundary; an across-null control caught it immediately, and the exact-zero propagation diagnostic then superseded the estimator entirely. The headline needs no estimator at all, which is exactly why it is exact.

We will return to this zero twice: as the reason committed records are free (§5), and as the *internal* reading of a ceiling whose *external* reading closes the paper (§6).

3. Charge is topological: quantized winding and an integral Gauss law

Electric charge is discrete, conserved, and unscreenable, the cleanest case for the ontology’s boldest claim, that a physical constant can be a global relational fact. This section asks, graph-first, whether the ED participation graph carries a topological invariant with charge’s skeletal features, forced by the graph structure rather than fit to it. It does. It also, just as sharply, does not produce a Coulomb field at the discrete layer, and we locate exactly where that field lives.

The discipline, stated first. The result is reached graph-first, never charge-first: the participation-graph structure decides which invariants exist, and only then are they compared to charge. No topology is retrofit to hit ± 1 or $\mp \frac{1}{3}$; those integers never appear in any result. What appears is an integer winding with *no selected magnitude*. Why the $U(1)$ holonomy in particular: among the invariants the graph admits, it is the one whose fundamental group $\pi_1(U(1)) = \mathbb{Z}$ makes its holonomy automatically integer-quantized, the natural candidate for a discrete conserved quantity, chosen for that reason and not because it resembles charge.

Quantize: an integer winding, conserved and protected.

On a $U(1)$ -on-a-cycle arena, single-valued committed polarity (P09 supplies the angle; P11 makes it single-valued) forces the holonomy around the cycle to an integer winding $w = (1/2\pi)\Sigma_e \Delta\varphi_e \in \mathbb{Z}$, confirmed to machine precision. The winding is conserved, and changing it discretely would require “uncommitting” a polarity, which P11 forbids (Paper B4).

Tier: the integer quantization is Measured ($\pi_1(U(1)) = \mathbb{Z}$, machine precision); the

conservation and irreversibility-protection are Derived (P11). This is the section’s anchor, and it is a concrete answer to a question standard physics leaves as a fact to be assumed, *why is charge quantized?* Here the answer is structural: the committed phase must return to itself around a loop, P11 makes that return single-valued, and the topology of the circle makes the number of wraps an integer that cannot change without an uncommitting the substrate forbids. Charge quantization is protected by the arrow.

Couple and source: inert, but obeying a Gauss law. Two immediate features follow, both consequences of orientation-blind selection. First, the winding is *inert* at the selection level, Σ never reads the phase, so the winding couples only weakly, through a discrete w -indexed bandwidth ladder $\propto \cos^2(\pi w/N)$ (P04, measured). It is a conserved topological invariant, not a dynamically active charge. Second, and more strikingly, it obeys the **integral content of Gauss’s law**: for any simple closed loop L enclosing the winding,

$$\Sigma_{e \in L} \Delta \varphi_e = 2\pi w,$$

the *same* integer w for every enclosing loop, independent of the loop’s size and shape (measured). This is the unscreenable, topological statement of charge-sourcing, the flux through any surface enclosing the winding fixed by w alone. It is genuine and graph-native.

The honest open edge: the metric flesh is one layer up. What is *not* present at the discrete layer is a determined local field, a per-edge Coulomb $1/r^2$ configuration. The circulation is fixed and topological, but how it distributes over individual edges is undetermined at the substrate, and this is not a gap to paper over. Relaxing orientation-blindness alone yields a sweep-dependent seam, not a field; forcing a determined isotropic $1/r^2$ field requires removing P11 as well, at which point the system is ordinary lattice-field relaxation and no longer ED. So the determined field is blocked by *both* invariants at once. Read as lattice gauge theory this is exactly right: the holonomies are link variables, the individual per-edge values are gauge-redundant, and the Coulomb field is the *continuum* limit of the link variables, not a substrate object. The relevant action is even latent in the coherence term, $\cos^2(\Delta\varphi/2) \approx 1 - \frac{1}{4}(\nabla\varphi)^2$, whose incoherence part is the standard $\int(\nabla\varphi)^2$ action density, but orientation-blind Σ does not minimize it at the micro layer. That converts a dead end into a well-posed open question, stated plainly and left open:

Does ED’s DCGT coarse-graining select the Maxwell-action configuration as the continuum expectation of the discrete holonomies? Unproven; **no Maxwell derivation is claimed** (Paper B4, §7).

This is the same open edge as the nonlinear metric in Part I, the determined continuum object as a coarse-grained shadow whose selection ED has not yet demonstrated. Charge is its cleanest instance, because the lattice-to-continuum dictionary is already standard, so the open question is unusually sharp: not “what continuum object should appear” but “does the coarse-graining measure select the known one.”

Complementarity, not deficiency. One reading ties the section together. The *same* two of the substrate’s commitments that realize the skeleton, orientation-blind (phase-blind) selection and irreversible commitment, are exactly what withhold the determined field, and they are the same commitments that make ED a determinacy engine in the first place. One architecture does both. The winding is quantized-but-inert, Gauss-law-sourcing-but-locally-undetermined, for a single reason, and that reason is not a limitation bolted on but the substrate’s defining

commitment.

What this section does not claim. ED does not derive electromagnetism, Maxwell’s equations, or the Coulomb field. It produces no charge magnitudes, no spectrum, no fine-structure constant; the integers ± 1 , $\mp \frac{1}{3}$ do not appear. The winding is inert and weakly coupled, and the correspondence to lattice gauge theory is structural, used only to locate the continuum field, not a claim that the substrate *is* a gauge theory.

4. Parity violation is necessarily spontaneous: the vector theorem

The weak force couples to one handedness. Does ED’s substrate produce that parity violation, or inherit it? The honest answer, sharpened here into a theorem, is that ED builds the entire *stage* of chirality and inherits the one *assignment*. The section grounds the chirality operator, proves the clean substrate carries no chirality, and names where the assignment actually comes from.

Ground the operator. The Dirac chirality operator is, by definition, $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$. Read its two factors in ED’s terms:

- γ^0 is **the arrow**: ED’s timelike direction is the commitment order (P11, retarded-only transport). γ^0 is the process primitive.
- $\gamma^1\gamma^2\gamma^3$ is **the spatial orientation**: the product of the three spatial gamma matrices is the oriented volume element, and the choice of $\gamma^1\gamma^2\gamma^3$ versus $\gamma^1\gamma^3\gamma^2$ is a handedness of the 3-frame, flipping sign under a spatial reflection, the parity-odd object it must be.

So $\gamma^5 = i \cdot (\text{the arrow}) \times (\text{the spatial orientation})$, both factors native (structural). This reconciles an earlier finding that ED carries no *local* per-channel “screw”: γ^5 is not a per-channel object at all, but a global one, the arrow times a single global orientation, which is why no local screw was ever found. Two properties make the operator physical. The orientation is **spontaneous** (Derived): ED’s rules are parity-symmetric, no primitive is a reflection, and a parity-symmetric rule set cannot fix a handedness, so any global orientation is chosen by spontaneous symmetry breaking, a ferromagnet picking a pole. And the emergent space is **orientable** within the causal patch (structural): ED has no reflection in its toolkit, so it cannot build an orientation-reversing holonomy, and growth from a single seed propagates one handedness across the patch. The same fact, no reflection primitive, does double duty.

The theorem. Model a family of N channels as a commitment map on an emergent chain, $H(k) = e^{ik}A + e^{-ik}B$, with A the forward (arrow-directed, retarded) cross-channel hop, B the backward hop, and k the emergent wavenumber. The chirality of such a map is its **point-gap winding**, the number of times $\det H(k)$ encircles a reference as k runs over its period; a nonzero winding is the net-chirality signature, zero is vector. ED’s substrate is parity-clean, so parity, the spatial reflection $k \rightarrow -k$ together with the lane reflection S of the channel row, is a symmetry, which forces the backward hop to mirror the forward hop:

$$B = SAS^{-1}, \text{ whence } SH(k)S^{-1} = H(-k), \text{ so } \det H(k) = \det H(-k).$$

The determinant is therefore an **even** function of k . As k runs over its period the determinant path traces out and retraces itself, encircling no reference point, so the point-gap winding is **identically zero, for every N and every forward hop A .**

The parity-clean substrate is vector for every channel-count. The clean substrate is vector for one channel, two channels, three channels, all of them (Paper CleanVector, §4).

Tier: Derived (theorem), at the channel-transport level. The mathematical step is short by design, an even determinant retraces its path; the load-bearing content is in the two inputs, that ED’s clean substrate is parity-symmetric of this commitment-map form, and that the point-gap winding is the operative chirality invariant. What the theorem adds over “a parity-symmetric object carries no parity-odd invariant” is that it holds for *every* channel-count with no exception, verified numerically as a control for $N = 1..6$, which is exactly the generality an earlier minimal-construction verdict lacked.

The corollary, the strong positive.

Since the clean transport is vector, any parity violation must break the parity symmetry. **If parity is violated in ED, its direction is necessarily spontaneous** (Derived corollary).

ED does not derive *that* parity is violated, that is the inherited assignment, below. It establishes that the direction, if there is one, cannot be a law-level handedness written into the rules, because the rules cannot carry one, but must be a contingent, symmetry-broken choice. A careful negative (“we did not find chirality where we looked”) has become a theorem (“the clean rules cannot carry it, and here is the proof”).

The casting is inherited. Which channel-count ends up chiral once the symmetry breaks is not fixed by the transport. The clean winding is 0; under a natural breaking the winding is N , monotone in the channel-count; the Standard Model’s actual pattern is *non-monotone*, one channel vector (electromagnetism), two chiral (the weak force), three vector (the strong force), matching neither. So *which* force is chiral is not a transport-winding fact. It lives in representation theory: the fundamental of $SU(2)$ is **pseudoreal** while the fundamentals of $SU(N \geq 3)$ are complex, an “ $N = 2$ is special” fact, not a parity-of- N fact. This ED **inherits**, exactly as it inherits the gauge-group multiplicities and the constants (account tier, the surviving candidate). ED builds the stage, inherits the one assignment.

Anomaly baseline. One corollary for the deepest consistency requirement of a chiral gauge theory. A gauge anomaly requires chiral content; a vector theory’s two handednesses cancel automatically. So the clean transport, being vector, is **automatically anomaly-free** at that level (structural, contingent on the descent to a genuine vector Dirac sector). The Standard Model’s *nontrivial* chiral cancellation, the balance of hypercharges tying quarks to leptons, is a property of the chiral content, which appears only after the breaking, and *which* content is the inherited casting, so it is inherited with the content, not independently derived.

The distinctive prediction, and its status. Because the clean substrate is provably vector, ED’s parity violation is necessarily a spontaneous, contingent, per-universe orientation, tied to the arrow, and the same arrow sets the matter/antimatter sign. So ED predicts the gauge handedness and the matter/antimatter sign are **correlated**, two faces of one first-commitment choice, against the Standard Model’s law-level, fixed, uncorrelated handedness (Prediction). This is the firmest footing the prediction has had, resting on a theorem rather than a stance. It is also **unconfirmed**, and no near-term measurement is in hand, this is the sector’s one forward bet and it has not been cashed. We flag it as squarely as Part II flagged the pressured mass

number: distinctive, principled, and open.

What this section does not claim. ED does not derive the weak force’s chirality (inherited). It produces no representation spectrum, no hypercharges, no masses or mixings. The theorem is at the channel-transport level; the relativistic substrate-to-Dirac descent, behind a fully relativistic chirality statement and the one non-inherited anomaly candidate, is the standing open arc.

5. Classical objectivity is unbounded: committed records are free

The three results so far are things the substrate forces about a single fact, its locality, its charge, its parity. This short section is about how *many* facts the classical world can hold, and the answer, a theorem, ties the sector back to the arrow.

The dictionary. Quantum Darwinism, the redundant imprinting of a system’s pointer-basis information into many environment fragments, the mechanism of classical objectivity, assembles inside ED from standing parts with no new machinery: the pointer basis is the arrow-selected commitment basis (einselection is primitive, P11’s irreversibility selects which content can be recorded); a system is a chain with channel content; an environment fragment is a chain whose cross-chain correlation carries the system’s pointer value; a classical record is that correlation *after* P11 commitment, an irreversible, unerasable fact.

The accounting theorem.

A P11-committed record consumes no dynamical (V5) budget. It is recorded structure, not active kernel content; it persists with no sustaining channel (the A1 result of §2, correlations that survive at zero channel capacity; A1 runs on a V5-free substrate, so it evidences channel-free persistence, not a V5-weight measurement), and it cannot be erased to refund anything (P11). Therefore the substrate’s finite dynamical budget bounds correlation *formation and coherent maintenance*, never the *stock* of committed records (Paper Darwinism, §2–§3).

Tier: Derived-structural. The consequence is a clean win, and it is the point of the section: accumulated classical redundancy is **unbounded**. Everyday objectivity at the magnitudes decoherence theory measures, on the order of 10^8 imprints per microsecond for a sunlit dust grain, is not merely tolerated by ED but *explained*. Under a naive reading in which the dynamical budget caps all cross-chain correlation, ED would be falsified by daylight; the theorem turns that around. Records are free precisely because they are committed, and the classical world is cheap for a principled reason: **the classical is the committed**. The same P11 that gives the arrow (§1), protects the charge winding (§3), and grounds γ^5 (§4) is what makes objectivity unbounded. One primitive, wearing every hat in this paper.

The fence. ED does say more about the *live* (uncommitted) side, a per-locus cap on simultaneously uncommitted fragments, and, under a further postulate, a ceiling on coherent branching-state width commensurate with a quantum-error-correction plateau. That further step rests on a **declared postulate** (P-QD-LiveWeight), consistent with the corpus’s budget accounting but not forced by it, so we name it and stop. This section features the accounting theorem and its unbounded-objectivity corollary, which stand on P11 alone. The width-ceiling weapon is

postulate-conditional and is not banked here.

6. The one limit, proven: the finite-memory ceiling

Every result so far is something the finite, committed substrate *forces*. The paper closes on what it *cannot* do, and this negative is unlike the program's others in one decisive respect: its "cannot" is a theorem, anchored to mathematics outside ED. It is also, in the end, the same fact as §2.

The class, by identity. ED's substrate is finite-reach and finite-memory by its primitives: every determination is a function of a bounded local state, which is a function of a bounded residue state $\mathbf{n} \bmod \mathbf{M}$, which is exactly a **periodic, zero-entropy sequence**. This is not an analogy but an identity, and it places the substrate squarely in the class that a century of analytic number theory has charted, the class the sieve methods and the Möbius-randomness program constrain.

The ruler. Use the integers as an external ruler, through the structure of their prime factorization. That structure splits, for a finite-local observer, into two layers. The **template** is what a finite residue state determines, which positions are definitely composite, and the equidistribution of primes among the residues left open, finite-local and representable by the substrate class. The **escape** is what no finite residue state determines, primality itself, the sign of the Möbius function, the pair correlations of twins, the parity barrier. External mathematics predicts a finite-memory class captures the first and provably not the second. We measure both.

The template is reproduced exactly. The scale-invariant template information $H(dx) - H(dx \mid n \bmod 30) = 1.700$ bits is matched exactly, and the least-prime-factor density ladder $P(lpf = p) = (1/p) \prod_{q < p} (1 - 1/q)$ is reproduced to five decimals (Measured). The finite-local layer is fully within the substrate class.

The escape is provably blocked, three ways. (A) Conditioning primality on any finite template state leaves an irreducible residual, $H(\text{prime} \mid n \bmod M) \approx 0.18\text{--}0.29$ bits per integer, that does not fall toward zero as the memory grows, the parity barrier expressed in bits, irreducible because Dirichlet's theorem makes the primes equidistribute among the open residues, so a finite residue state cannot tell which open position is prime. (B), the sharp test, the *optimal* finite-memory correlator with the Möbius function, $C * (N, M) = (1/N) \sum_r |\sum_{n \equiv r(M)} \mu(n)|$, decays toward zero as N grows (bounded by $\sqrt{M/N}$) for every fixed M ; even the best function of the template state cannot correlate with μ asymptotically. This is the paper's one genuine experiment, a bounded-complexity function with $\mathbf{C}^*$ bounded away from zero would be a real mathematical discovery, and it **was searched for, over all functions of $\mathbf{n} \bmod \mathbf{M}$, and not seen**. (C) The twin-prime density carries the Hardy–Littlewood singular series $2C_2 = 1.3203$, a pair-correlation a finite-local template generator does not supply. **Tier: Measured, and anchored to external theorems.**

The mechanism. Why does no *fixed* finite memory ever reach the escape? Because the template state that would resolve more of the structure grows with the scale it must resolve, on the order of $\sqrt{N}$, the same square-root-cancellation scaling that governs the large sieve. A larger memory does correlate with μ at small N , but the correlation washes out as N grows.

The escape recedes at exactly the rate that keeps it out of finite reach, the $\sqrt{N}$ **activation horizon**, the substrate-level shadow of the parity barrier.

Why this is the program’s one proven negative. Most of the corpus’s negatives are coarse-graining walls, an obstruction identified but not proven insurmountable. This one is different, because the wall is a **theorem** external to ED: Möbius orthogonality to *periodic* sequences is classical ($\sum_{n \equiv r(M)} \mu(n) = o(N)$ by the prime number theorem in arithmetic progressions, Siegel–Walfisz), and the special case the substrate needs is that already-proven base case, not Sarnak’s open conjectural generalization. The sieve parity barrier (Selberg) is the same limit stated sieve-theoretically. ED is not grading its own homework; the “no” is read off mathematics that stands independent of it. That external anchor is exactly what upgrades this from a located obstruction to a proven one.

The closing identity. And here the paper’s two ends meet. The channel-capacity result of §2 measured ED’s ceiling *inside*, against its own observables, and found controlled capacity exactly zero. The finite-memory result measures the *same* finite-reach ceiling against an external, math-certified ruler, and finds the escape provably beyond it. These are not two ceilings but one, read twice: the finiteness that made a committed fact strictly local (§2) is the finiteness that bounds what the substrate can ever compute (§6). Exact locality opened the family of things the substrate forces; the proven limit closes it. **What this section does not claim:** ED does not generate or explain the primes; they are a ruler, not a target, and no result in number theory is advanced. The asymptotic statements are anchored to the proven theorems and read from finite- N trends.

7. What this sector does not claim

The results above are firmer in kind than the forms of Parts I and II, and for that reason the disclaimers matter more, not less. Stated plainly:

- **No confirmed prediction.** This is the first and most important. Unlike the gravity sector, which ended on a confirmed evolving acceleration scale, this sector has no empirical scalp. Its one distinctive forward bet, the spontaneous parity handedness correlated with the matter/antimatter sign (§4), is unconfirmed and not near-term testable. The sector’s currency is derivation purity, not observational contact, and that is a real limitation.
- **Locality (§2)** is an exact zero on the tested cases, one certified substrate and five baselines with a passing control, not a geometry-universal proof; and it is confirmed empirically on the substrate, not derived from the contrast-first ontology.
- **The “locality without superdeterminism” reading (§2)** is a cross-sector synthesis, not an A1 result. A1 supplies only the exact no-signaling (capacity zero); the Bell–EPR correlations it makes *local* are reproduced in the entanglement sector (Part II), which inherits the Tsirelson bound and adds no new bound. The reading depends on Part II and is interpretive, not a theorem of this sector.
- **Charge (§3)** produces no magnitudes, no spectrum, no fine-structure constant, and no Maxwell or Coulomb field at the discrete layer; whether DCGT coarse-graining yields Maxwell/Coulomb in the continuum limit is an honest **open** question, unproven, with no derivation claimed.
- **Parity (§4):** the theorem is that the *clean* substrate is vector, at the channel-transport

level; *which* force is chiral (the casting) is **inherited**, representation-theoretic. The relativistic substrate-to-Dirac descent and the full chiral anomaly cancellation are open.

- **Objectivity (§5):** only the accounting theorem and unbounded objectivity are featured, standing on P11 alone. The live-redundancy width ceiling and its commensurability with the error-correction plateau rest on a **declared postulate** (P-QD-LiveWeight) and are not banked here.
- **The limit (§6)** is a ceiling-location, not a generation. ED does not generate the primes; the external theorems (Siegel–Walfisz, the sieve parity barrier) are what make the “no” proven, and the asymptotics are read from finite- N trends against those theorems.
- **Nothing is forced from nothing.** Every result is conditional on the thirteen primitives and, in particular, on the three architectural facts, the substrate’s finiteness, its cleanness, and its irreversible commitment, which are themselves postulated, not derived. The claim is that, granted them, this family follows as theorems, one of which is a proven limit.

8. Appendix: tier table

Every load-bearing claim, with its tier transcribed from the substrate-evaluation tiered-claims ledger and verified against the source paper.

Result	Claim	Tier	Inherited / open
A1	Controlled channel capacity between distinct regions = 0	Derived (from exact measured $L2 = 0$)	locality confirmed empirically, not derived from the ontology; one substrate, 5/5
A1	Distant-region propagation exactly zero ($L2 = 0.000000$)	Measured (exact)	not a geometry-universal proof
A1	Boundary = common-cause, not channel; locality without superdeterminism	Interpretation (on the measured zero)	—
B4	Quantized integer winding $w \in \mathbb{Z}$ ($\pi_1(U(1)) = \mathbb{Z}$)	Measured (machine precision)	no selected magnitude; $\pm 1, \mp \frac{1}{3}$ not produced
B4	Winding conserved, irreversibility-protected	Derived (P11 forbids uncommitting)	—
B4	Integral Gauss law, circulation = $2\pi w$, loop-independent	Measured / Grounded	local Coulomb field withheld (open)
B4	Maxwell action latent in $\cos^2(\Delta\varphi/2)$	D-structural	DCGT $\rightarrow$ Maxwell/Coulomb OPEN , unproven

Result	Claim	Tier	Inherited / open
CleanVector	$\gamma^5 = \text{arrow} \times$ spatial orientation	Structural	contingent on substrate→Dirac descent
CleanVector	Orientation spontaneous	Derived	—
CleanVector	Clean substrate is vector for every N ($\det H(k)$ even $\implies$ winding $\equiv 0$)	Derived (theorem)	transport-level; Dirac descent open
CleanVector	If parity violated, direction necessarily spontaneous	Derived (corollary)	<i>that</i> parity is violated is inherited
CleanVector	Casting (which force is chiral) = pseudoreality of $SU(2)$	Account (surviving candidate)	inherited , representation- theoretic
CleanVector	Clean baseline anomaly-free; chiral cancellation inherited	Structural + Inherited	contingent on Dirac descent
CleanVector	Parity handedness correlated with matter/antimatter sign	Prediction	unconfirmed , no near-term test
Darwinism	Committed record costs no dynamical budget $\implies$ objectivity unbounded	Derived-structural	absolute budget scale inherited
Darwinism	GHZ-width ceiling / plateau commensurability	Postulated (declared), P-QD-LiveWeight	not banked here
Primes	Substrate is finite-reach / finite-memory class	Derived (primitives #2-#3)	—
Primes	Prime <i>template</i> reproduced (1.700-bit invariant; lpf ladder)	Measured	—
Primes	<i>Escape</i> provably blocked (parity barrier; Möbius correlator $\rightarrow 0$; twin $2C_2$)	Measured + externally anchored	Siegel–Walfisz; sieve parity barrier
Primes	$\sqrt{N}$ activation horizon	D-structural	—

Result	Claim	Tier	Inherited / open
Primes	The “no” is the program’s only <i>proven</i> negative	Interpretation	anchored to external math, not ED grading

Series context. Part III of the Selected Formalisms series, after gravity (I) and quantum fields (II). Where those sectors reproduce a law’s form and inherit its magnitude, this sector collects the results the committed substrate forces as theorems, locality (A1), topological charge (B4), the vector/parity theorem (CleanVector), and unbounded objectivity (Darwinism), all consequences of three architectural facts, the substrate’s finiteness, its cleanness, and its irreversible commitment, and closes on the one limit the same finiteness makes proven, the finite-memory ceiling (Primes). The sector’s price is stated as plainly as its theorems: no confirmed prediction, derivation purity in its place.